

How Dynamic Inflow Survives in the Competitive World of Rotorcraft Aerodynamics

The Alexander Nikolsky Honorary Lecture



David A. Peters*

*McDonnell Douglas Professor of Engineering**Department of Mechanical, Aerospace, and Structural Engineering, Washington University
St. Louis, MO*

Dynamic wake models have found a firm place in rotary-wing analysis since their inception in the 1950s, proliferation in the 1970s, and maturation in the 1990s. They have maintained their usefulness, despite the appearance of new and more powerful tools—such as prescribed-wake models, free-wake models, and computational fluid dynamics—due to the following five fundamental reasons: (1) the various models and improvements have always come in response to important, yet unexplained experimental results; (2) the response to those results was invariably based on sound physical intuition as to the nature of the discrepancies; (3) the model improvements at each step were based on engineering physics rather than heuristic fit of data; (4) the models included only enough physics to explain the phenomenon—and no more; and (5) each model was hierarchical to earlier models so that no model was ever replaced—i.e., each new improvement included earlier models as special cases. Because of this, dynamic wake models have maintained a strength in the domains of real-time flight simulation, stability computations, and flight mechanics and control. This paper looks in detail at how these developments have transpired and how they relate to the importance of simplified tools, in general.

Nomenclature

a	slope of lift curve, rad^{-1}
a^*	equivalent lift-curve slope
C_L, C_M	roll-moment and pitch-moment coefficients
C_T	thrust coefficient
$[C]$	coupling matrix of dynamic wake models
c	blade chord, m
$\{F\}$	forcing functions of dynamic wake models
h	rotor height above ground divided by R
I_y	blade inertia, $\text{kg}\cdot\text{m}^2$
K_G	ground effect factor
K_{Re}	wake skew intensity parameter
$[L]$	matrix of influence coefficients
$[M]$	mass matrix of dynamic wake models
N	number of blades
p	blade flapping frequency, per/rev
q_b	pitch rate
R	blade radius, m
r	nondimensional radial coordinate
T	nondimensional time constant
t	time, s

V	mass-flow parameter
X	wake skew parameter = $\tan(\chi/2)$
β	blade flapping angle, rad
γ, γ^*	blade Lock number, equivalent Lock number
ζ	blade lag angle, rad
θ	body pitch angle, rad
$\theta_0, \theta_s, \theta_c$	collective and cyclic pitch, rad
κ	wake curvature
λ	climb ratio
μ	advance ratio
μ_0	propeller advance, $1/\eta$
v	average induced flow ratio
v_n	states of dynamic wake models
v_0, v_s, v_c	states of Pitt model
ρ	density of air, kg/m^3
σ	blade solidity
τ	nondimensional time, $\Omega t, ()^* = d/d\tau$
ν	total dynamic inflow
ϕ	body roll angle, rad
χ	wake skew angle, rad
ψ	azimuth angle, rad
Ω	rotor speed, rad/s
ω	frequency of oscillation, per/rev or Hz
$()_R, ()_P$	regressing, progressing mode

*Corresponding author: email: dap@me.wustl.edu. Received June 2008; accepted November 2008.

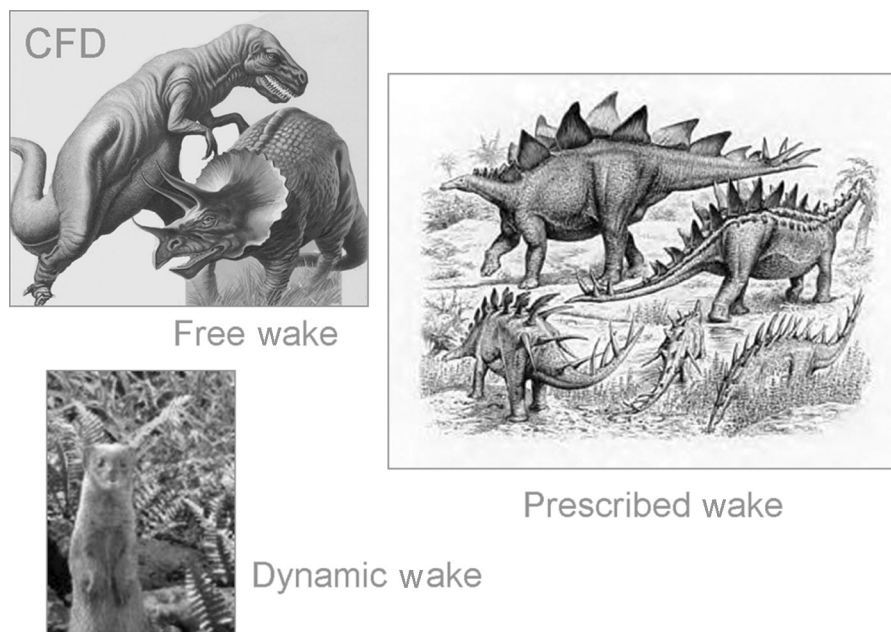


Fig. 1. Ecological niches of aerodynamic models.

Introduction

In the very competitive world of aerodynamic modeling of rotorcraft, one could argue that computational fluid dynamics (CFD) is the top predator (Fig. 1), and that it fills many competitive niches. However, CFD is not well suited to the computation of the induced flow due to wakes because the artificial viscosity of those codes (necessary for numerical stability) tends to cause concentrated vorticity to decay, thus losing the most powerful source of induced flow—the vortex wake. Thus, free-wake models (rather than CFD) tend to dominate the tools used to find induced flow fields for rotorcraft.

Free-wake models, however, are not perfect. They rely on many convergence parameters such as vortex-core size, number of radial and azimuthal filaments, and the number of wake turns used before truncation. In addition, the models are fairly unstable at low speeds and completely unstable in hover. To enhance numerical stability, free-wake models are often replaced by prescribed-wake models as the method of choice to find the induced flow field; although prescribed-wake models still have the other convergence issues. Nevertheless, the utility of these vortex-lattice codes far exceeds their drawbacks, and they are used extensively in the industry.

Despite the existence of these very attractive and powerful tools (CFD and vortex-lattice), dynamic wake models have persisted as being the model of choice in many rotary-wing applications and especially in dynamics, stability and control, and real-time flight simulation. The purpose of this paper is to look back through the history of the development of dynamic inflow models and try to determine why they have been so successful and why they continue to be used in many applications. Along the way, it will also be noted that many Nikolsky lecturers have been involved in the development of dynamic inflow models; see Fig. 2.

To begin, we should define what we mean by a dynamic wake (i.e., dynamic inflow) model. I believe that there are three characteristics that set a theory aside as being in this genre. First, it must be a mathematical model that, given the time history of blade loads on a rotor, will produce the induced flow normal to the rotor disk as a function of the key coordinates: time, radial position, and azimuthal position. Thus, it is a theory that separates the wake model from both the airloads and structural models in a formal way (i.e., it asks neither how the time history of loads

has been computed nor for what purpose they will be used). Second, the form of a dynamic wake model must be in terms of a finite number of ordinary differential equations in time that take the form

$$[M] \left\{ \frac{dv_n}{dt} \right\} + [C]\{v_n\} = \{F_m\} \quad (1)$$

where the v_n are states that define the flow field, $[M]$ is an apparent mass matrix, $[C]$ is a matrix of influence coefficients, and $\{F_m\}$ depends on the blade loading. It follows that the v_n are state variables for the flow field that are driven by the blade loads. Third, a dynamic wake model must be formulated in such a way that the number of states chosen can be determined by the user based on the fidelity of the model needed for a given application. It must be a hierarchical model in which one might use a single state, three states—or even more—depending upon the application. This defines what we discuss here as dynamic wake (or dynamic inflow) models.

Foundation (1950–1969)

Simple momentum theory

The origins of dynamic wake models go back to the work of Ken Amer (the 1988 Nikolsky Lecturer) when he was examining some experimental data on the roll damping of helicopters (Ref. 1). Amer noted that the measured roll damping was significantly larger than that predicted by theory, and this brings us to our first point about dynamic wake models through the years. They tend to follow directly from unexplained experimental data that have a significant impact on helicopter design and understanding. This was the case with Amer, and it has remained the case throughout the development of dynamic inflow models. Experimental results have consistently motivated the development of improvements to the theory.

Returning to Amer, it is noteworthy that he offered a plausible explanation of the unexplained data. On page 1 of Ref. 1, Amer writes:

The . . . discrepancy between the data and the theory appears to be due primarily to the changes in the distribution of thrust around the rotor disk. These changes in induced velocity are



Wieslaw Stepniewski - 1981



Rene Miller - 1983



Bob Loewy - 1984



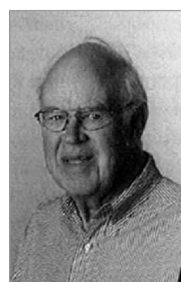
Ken Amer - 1988



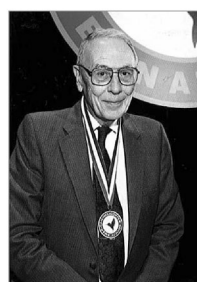
Robin Gray - 1991



Dan Schrage - 1999



Pat Curtiss - 2000



Frank Harris - 2006



David Peters - 2008

Fig. 2.

not taken into account in the theoretical calculations because of the excessive labor that would be involved.

Amer was exactly correct in this conjecture, and this brings us to our second point on the effectiveness of dynamic wake models. The development of dynamic wake models has been based on keen physical insight as to what could be creating the discrepancy. In this case, Amer's intuition took him straight to the heart of the phenomenon.

In particular, the roll damping of a helicopter arises from the tilt of the tip-path plane that develops in response to a roll rate. The resultant tilt of the thrust vector tends to oppose the roll motion, thus giving a roll damping. The tilt is due to the blade flapping response to the gyroscopic and aerodynamic forces created by roll rate. Since these forces are near the natural frequency of the blades (once per rev), the tilt is inversely proportional to blade damping. Thus, since blade damping is $\gamma/16$, it follows that the roll damping is $16/\gamma$, where γ is the Lock number.

The Lock number is a ratio of aerodynamic forces to inertial forces in a helicopter,

$$\gamma = \rho a c R^4 / I_y \quad (2)$$

where ρ is the density of air, a is the slope of the lift curve, c is the blade chord, R is the blade radius, and I_y is the blade mass moment of inertia about the flapping hinge. The change in thrust from side to side on the rotor creates a change in inflow from side to side that, in turn,

lowers the differential in angle of attack and decreases the aerodynamic effectiveness of the blades—often by as much as 50%. Thus, blade damping goes down and roll damping goes up. Amer saw that this must be the case.

Sissingh (Ref. 2) soon showed that the computation effort was not so prohibitive as Amer had originally thought, and Sissingh derived inflow formulas for both hover and forward flight. Sissingh used a very simple, straightforward model to predict this effect. He applied standard momentum theory to the rotor slipstream but in a new way. He treated the two sides of the rotor disk as separate momentum tubes, each with a different thrust and, consequently, each with a different induced flow. From this, he could predict in closed form the relationship between the aerodynamic roll moment and the gradient in flow. His formulas explained the discrepancies noted by Sissingh in Ref. 2. This was the first dynamic inflow model, albeit quasi-steady.

The manner in which Sissingh performed his analysis brings us to the third and fourth important points about the success of dynamic wake models. First, they are based on engineering physics, and, second, they include just enough physics to explain the effect—but no more. In this case, Sissingh used a simple momentum theory on each half of the rotor disk. This was all that was necessary to obtain what he needed. The flow was assumed constant with radius (implying a discontinuity across the center), but that did not matter. No further refinement was needed to explain the Amer data. The fact that dynamic wake models are based on

physics has given them both staying power and growing power. Had Amer and Sissingh rather decided simply to put an empirical correction factor on roll damping, future models could not have developed from it. The fact that Sissingh derived the correction factor from momentum theory gave the model a physical basis and a place from which to expand. Since the theory was made as simple as possible, it could both give physical insight and become a useful tool—applied over a broad range of problems.

Vortex considerations

Sissingh's work was well known in Great Britain but did not make deep inroads into developments in the United States for a number of years. Meanwhile, however, other researchers in America were coming to similarly interesting results. Once again, these results were motivated by unexplained phenomena. It was noted that, at zero or low lift, a helicopter rotor could develop a flutter when the natural frequency of the blades was near an integer multiple of the number of blades times the rotor speed, $N\Omega$. The effect was most pronounced at low values of lift coefficient. Robert Loewy (the 1984 Nikolsky Lecturer) conjectured that the effect might be due to the layers of returning vorticity beneath each blade (Ref. 3). When the natural frequency was near an integer multiple of $N\Omega$, these vortices might line up, creating an increased inflow and loss of aerodynamic damping.

As with the Amer conjecture, Loewy's explanation turned out to be exactly correct. He worked out a lift-deficiency function for the approximation in which the helical, returning wake is represented by a two-dimensional system with infinite layers of vorticity spaced below the airfoil. It turns out that the physics of this hypothesis was all that was needed to completely explain the phenomenon. The Loewy lift-deficiency function showed a large drop in damping at the precise integer multiples where the experimental effect had been noticed, and the drop was inversely proportional to the wake spacing (Fig. 3). Thus, damping went to zero as thrust went to zero, just as in the data. An interesting spin-off of the Loewy results is that they show—for cyclic pitch inputs (such as are typical of helicopter pitch or roll)—that even the static lift ($\omega = 0$ is an integer multiple of $N\Omega$) is lowered by this large factor. The loss of lift for static pitch or roll (at $\omega = 0$) is identical to the loss predicted by Sissingh from momentum theory. However, the Loewy theory went beyond Sissingh in that it treated other frequencies. The Loewy result also showed that, for unsteady inputs removed from $N\Omega$, there would be a time delay (i.e., a magnitude and phase shift) from the static value. Thus, Loewy theory was not simply quasi-steady, it was unsteady. Although the Loewy theory was in the frequency domain—and therefore not a true dynamic wake model—it contained the seeds of dynamic wake models to come.

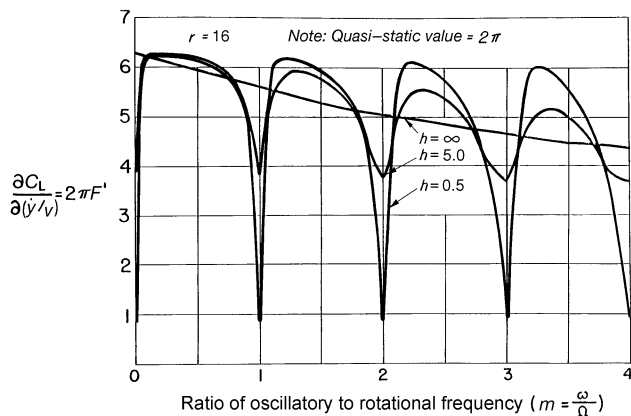


Fig. 3. Flap damping based on Loewy lift deficiency.

As an interesting sidelight, Rene Miller (the 1983 Nikolsky Lecturer) added some three-dimensional corrections to Loewy theory (Ref. 4), but these additions never had the popularity of the original Loewy function. The reason for this is that Loewy had just enough physics to explain the phenomena—and no more. Another interesting aspect of the Loewy results in Fig. 3 is a comparison of the various damping curves with the curve for $h = \infty$, which is—in essence—Theodorsen theory. Although the damping based on Theodorsen monotonically decreases with frequency, the more accurate Loewy theory shows that—away from integer multiples—the lift deficiency remains close to unity due to the time delay terms. In other words, the major effect of the returning wake is to cancel most of the Theodorsen effects and to replace them with Sissingh effects. Thus, for rotor blades, it is better to use no unsteady aerodynamics at all than to use Theodorsen theory.

Development (1970–1989)

Rediscovery

The next two decades, the 1970s and 1980s, saw the development and popularization of dynamic inflow and other related dynamic wake models. In 1971, Pat Curtiss (the 2000 Nikolsky winner) and Norm Shupe showed that the Sissingh theory could be greatly simplified and treated as an equivalent Lock number (Ref. 5). In particular, the Curtiss derivation showed that the net effect of the Sissingh equations for a static input is an equivalent reduction in lift-curve slope leading to a reduced value a^* implying a reduction in Lock number (see Eq. (2))

$$a^*/a = \gamma^*/\gamma = [1 + \sigma a/8V]^{-1} \quad (3)$$

where σ is the blade solidity and V is the nondimensional mass flow parameter. Curtiss pointed out that this drop in effective lift agrees with the Loewy lift-deficiency function for cyclic modes at integer-multiple frequencies. Curtiss further recognized that, for time-varying cyclic inputs, there must be a time lag for the buildup of inflow (and, correspondingly, for the reduction in lift). Thus, Curtiss certainly stands as the father of dynamic inflow modeling.

Another interesting development during this era was that Sissingh had moved from England to the United States and had begun to work for Lockheed California in the late 1960s. He—along with another British engineer, Michael Harrison—was working on the development of REXOR, the first attempt at a comprehensive helicopter code. Harrison promptly put the Sissingh inflow effect into REXOR—including an inflow time delay based on the mass of an assumed height of a column of air at the disk (Ref. 6) (Appendix A). Thus, by 1970 the roots of dynamic inflow modeling had found their way into rotor analysis in the United States through two different paths: (1) Pat Curtiss at Princeton and (2) Michael Harrison at Lockheed. (The REXOR code and theory manual were not published openly until the late 1970s (Ref. 6).)

Meanwhile, in 1970 the U.S. Army had set up the Army Aeronautical Research Laboratory (AARL) under the direction of Paul Yaggy. AARL was colocated with NASA Ames at Moffett Field, California. Bob Ormiston had come to lead the Army dynamics group, and he hired a young engineer from Washington University, Dave Peters. At that time, numerous data on the response of hingeless rotors to pitch and shaft inputs were being generated from wind tunnel tests in the NASA 40×80 and 7×10 low-speed wind tunnels. The motivation for these tests arose from the AH-56 Cheyenne program. The AH-56 was Lockheed's hingeless-rotor, compound helicopter. The challenges of that program led the Army to form the AARL (later called AMRDL, the Army Air Mobility Research and Development Laboratory) to look to the science of helicopter dynamics.

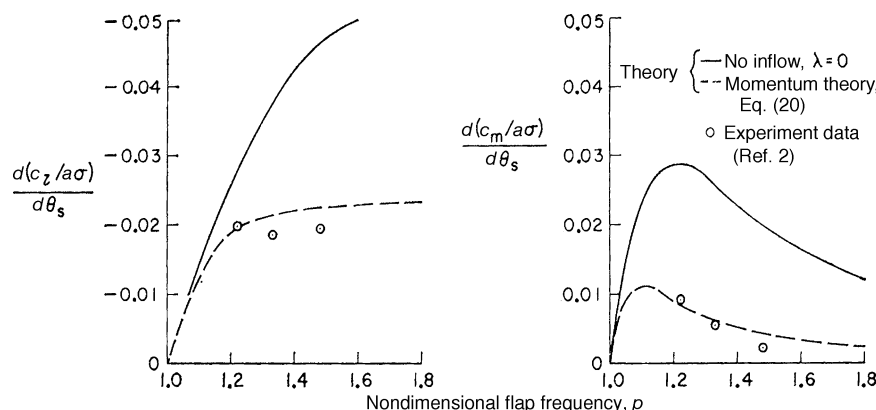


Fig. 4. Response data from 7.5-ft model rotor, $\gamma = 5.0$, $\sigma a = 0.729$.

It was clear from the NASA/Army data that the existing theories were not adequate to explain the test results, and Bob Ormiston was looking for answers. He instructed Dave Peters to write a specialized code that could compute rotor response, including all of the usual suspects for lack of correlation: elastic blade bending, reversed flow, root cutout, tip loss, and higher harmonics. Despite these efforts, the inclusion of these effects failed to explain the discrepancies. However, having come from Princeton and having studied with Pat Curtiss, Ormiston was aware of the dynamic wake effect of Sissingh. Fortunately, the Peters codes were perfectly structured for adding this type of wake effect in a rigorous manner.

Immediately upon addition of the Sissingh terms, the calculations for static data in hover agreed with the experimental data as shown in Fig. 4, taken from Ref. 8. In the results of Fig. 4, the rotor is slowed, thus giving a range of flapping frequencies, p . One can see that the traditional theories are off by a factor greater than 2 in the response predictions of roll and pitch moments due to cyclic pitch. The theory of Ormiston and Peters (with the Sissingh terms), on the other hand, nicely matches the experimental data. It is clear from the results in Fig. 4 that the quasi-steady dynamic wake model is able to predict most of the important effects in these static hover tests. (Interestingly, because REXOR was so expensive to run, it was not practical for Lockheed to use it for these comparisons. Otherwise, that code (with the Sissingh model) would have successfully predicted the experimental results.)

Despite this success in steady hover conditions, the use of the Sissingh inflow equations was unable to provide adequate correlation for either dynamic data in hover, static data in forward flight, or dynamic data in forward flight. Ormiston, like Michael Harrison and Pat Curtiss before him, had the idea of adding some type of time delay to the inflow equations. However, rather than a heuristic value of time delay—as had been applied in earlier work—Ormiston decided that there should be an analytical determination of this parameter. He knew about a study in the 1950s by Carpenter and Fridovitch in which they applied step and ramp inputs for collective pitch and then observed the buildup of inflow (and the resultant loss of lift) as a function of time (Ref. 7). Carpenter and Fridovitch had found that the apparent mass of an impermeable disk (being accelerated in still air) gave the appropriate time delay for the development of collective inflow mode.

Ormiston suggested to Peters that he use both the apparent mass and the apparent inertia of an impermeable disk for the time delays of the inflow to model this effect for both collective and cyclic modes. The addition of these apparent mass terms resulted in a set of dynamic wake equations for hover based entirely on physics.

$$[8/(3\pi)]dv_0/d\tau + [2V]v_0 = C_T \quad (4)$$

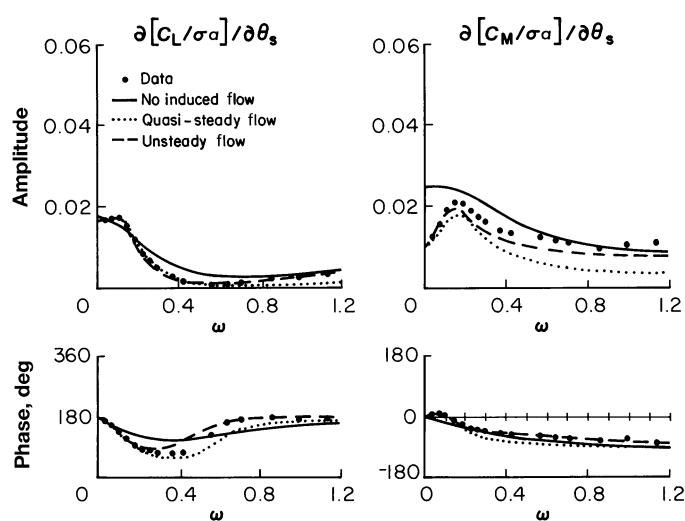


Fig. 5. Rotor dynamic response to cyclic pitch, $p = 1.15$, $\gamma = 4.25$.

$$[16/(45\pi)]dv_s/d\tau + [V/2]v_s = -C_L \quad (5)$$

$$[16/(45\pi)]dv_c/d\tau + [V/2]v_c = -C_M \quad (6)$$

where τ is nondimensional time, $\Omega\tau$; v_0 , v_s , and v_c are the uniform, side-to-side, and fore-to-aft gradients in induced flow; and V is the mass flow parameter, to be defined later. Results obtained with these equations provided excellent correlation with all of the steady and unsteady hover data taken by NASA and the Army, as can be seen in Figs. 5 and 6 taken from Ref. 9. At this same time, Dave Peters' former advisor Kurt Hohenemser (and his students back at Washington University) was running dynamic tests of their own and finding similar discrepancies as were occurring at Ames. They decided to use a dynamic wake model—similar to that of Eqs. (5) and (6)—but to use parameter identification to find the gains and time constants of that system (Ref. 10). Figure 7 shows the results of the Hohenemser investigation. In that test, one shaft spun the rotor while another turned the cyclic. Thus, pure regressing or progressing inputs could be given to the system based on the difference in the two shaft speeds. Hohenemser and his group determined that a gain of 2.0 and a time constant of 4.0 gave the optimally best correlation of the data. These values turn out to be within 2% of the values as taken from the analytic formulation of Eqs. (5) and (6).

At this point, the validity of the unsteady dynamic inflow model in hover was well established, but the correlations in forward flight were still very poor. It was clear that—for forward flight—the form of Eqs. (4)–(6)

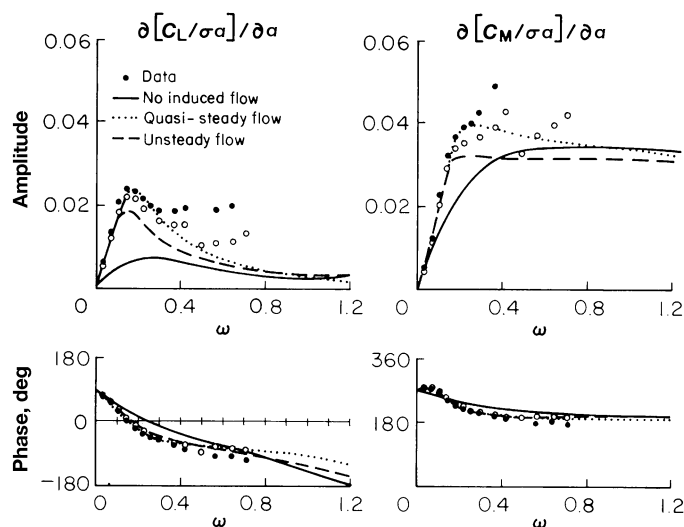


Fig. 6. Rotor dynamic response to shaft oscillations in hover.

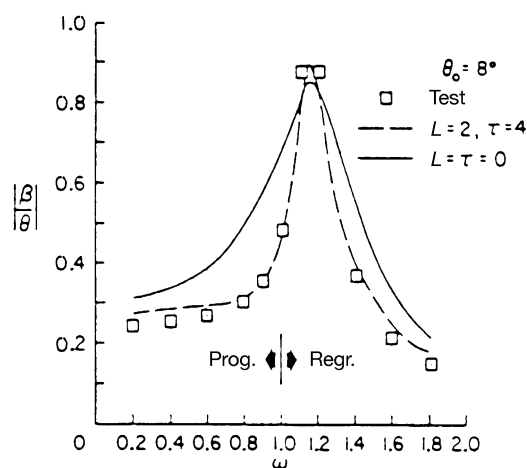


Fig. 7. Blade flapping response to rotating input.

would have to be modified to include couplings between the collective and cyclic modes. Ormiston and Peters assumed a 3×3 $[L]$ matrix to represent the couplings among the equations. They then attempted to determine the values of these couplings via a number of alternative approaches—including both modeling the rotor as a horseshoe vortex and applying pure parameter identification. Nothing, however, was entirely satisfactory (Ref. 9).

Skewed flow

During this same time period, computer power was growing, and some investigators like Jack Landgrebe were beginning to model the wake by vortex lattice methods either with a fixed lattice geometry or with a free wake (Ref. 11). Dave Peters returned to Washington University in 1975 and was convinced that what needed to be done was to exercise these vortex-lattice codes in controlled numerical experiments and thereby identify the coupling parameters of the system. Dale Pitt, an engineer at the Army Aviation Systems Command in St. Louis, came to Washington University as Peters' second doctoral student, and he had access to the Landgrebe code. Peters gave him the task of using the Landgrebe code to find $[L]$.

However, Pitt wanted to do a literature search first. Pitt won out over Peters' initial objections. He thoroughly searched the literature and found that Mangler and Squire (Ref. 12) had computed rotor-induced velocities from potential flow theory based on the circular wing theory of Kinnier (Ref. 13). (Interestingly, Stepniewski, the 1981 inaugural Nikolsky Lecturer, used the Mangler and Squire theory in a numerical approach to perform induced flow calculations (Ref. 14). He showed that the average induced flow from potential flow was the same as that of Glauert's momentum theory.)

Pitt used the potential functions of Mangler and Squire to develop a closed-form representation of the $[L]$ and $[M]$ matrices that would both couple the inflow equations and determine the time delays of Eq. (1) (Ref. 15). His theory is given below.

$$[M] \begin{Bmatrix} v_0^* \\ v_s \\ v_c \end{Bmatrix} + V[L]^{-1} \begin{Bmatrix} v_0 \\ v_s \\ v_c \end{Bmatrix} = \begin{Bmatrix} C_T \\ -C_L \\ -C_M \end{Bmatrix} \quad (7)$$

$$V = \frac{\mu^2 + (\lambda + v)(\lambda + 2v)}{\sqrt{\mu^2 + (\lambda + v)^2}} \quad (8)$$

$$[M] = \begin{bmatrix} \frac{8}{3\pi} & 0 & 0 \\ 0 & \frac{16}{45\pi} & 0 \\ 0 & 0 & \frac{16}{45\pi} \end{bmatrix} \quad (9)$$

$$[L] = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{15\pi}{64}X \\ 0 & 2(1 + X^2) & 0 \\ \frac{15\pi}{64}X & 0 & 2(1 - X^2) \end{bmatrix} \quad (10)$$

$$v = v_0 + v_s r \sin \psi + v_c r \cos \psi \quad (11)$$

Equation (7) is the standard form of the Pitt model with $(\cdot)^* = d(\cdot)/dt$. It includes the effect of wake contraction through the mass flow parameter V (Eq. (8)), which is written in terms of advance ratio μ , climb rate λ , and average induced flow v . The mass matrix (Eq. (9)) gives the same inertia terms as Peters developed based on the impermeable disk. The effect of wake skew angle χ appears explicitly in the parameter $X = \tan(\chi/2)$ (Eq. (10)). These wake skew terms are the important contribution by Pitt to the model. Once this set of equations is solved, the final induced flow is defined by Eq. (11) in which v is the total induced velocity at any point on the disk (normalized on tip speed), r is the nondimensional radial coordinate, and ψ is the azimuth angle.

As it turns out, Pitt eventually ran the Landgrebe code to identify $[L]$ and compared that with his closed-form $[L]$ matrix, as can be seen in Fig. 8. All elements show excellent correlation between the two approaches. However, because the Pitt model is in closed form based on potential flow theory, it provides a more applicable framework for future developments than does a model based on numerical fit of vortex-lattice results.

Once the Pitt theory was in place, it was straightforward to go back and correlate the original 7×10 data that had begun this whole line of investigation. Gopal Gaonkar did an excellent job of this (Ref. 16). Figure 9 shows typical examples of how the theory with no inflow, the theory with combined momentum theory and apparent mass (Eqs. (4)–(6)) and the Pitt model compare with the experimental data. The improvement in data correlation of computations with the Pitt model over computations with the previous theories is dramatic.

As these results were widely disseminated, the Pitt model systematically became the standard fare in many flight simulation and rotor

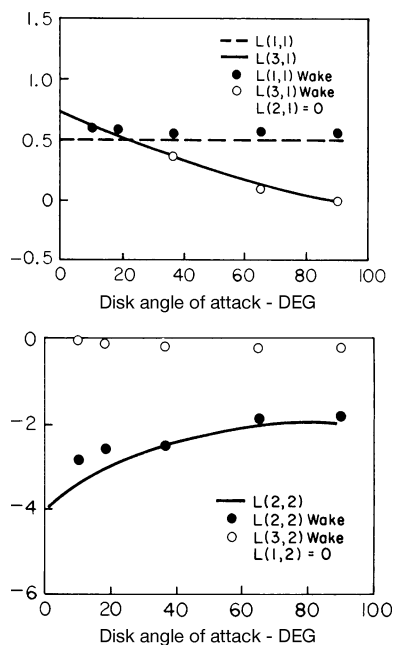


Fig. 8. $[L]$ matrix from the Landgrebe code compared with the Pitt model.

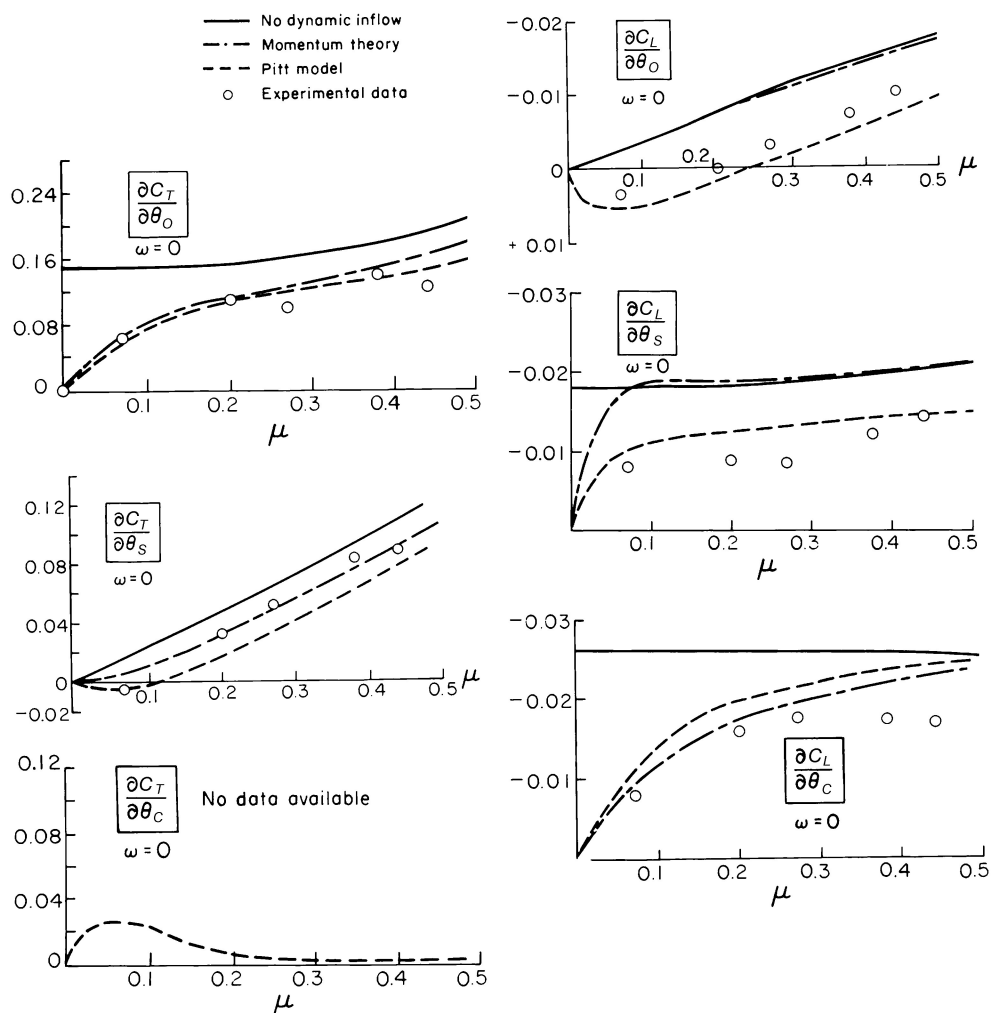


Fig. 9. Gaonkar results for rotor flapping response in forward flight.

response codes. Simultaneously, it was becoming clear that the effect of dynamic inflow would also be essential for coupled rotor-body stability analysis. This became evident as Bill Bousman began experiments on air and ground resonance with a dynamically scaled model.

Wayne Johnson (Ref. 17) had been using his new code CAMRAD to correlate this new rotor-body dynamics data taken by Bill Bousman. He soon discovered that not even the frequency calculations were accurate if dynamic wake models were not included. Figure 10 shows the improvement in frequency and damping when dynamic inflow is included in CAMRAD.

In Fig. 10, open symbols are the experimental data (i.e., frequencies extracted from test data by harmonic analysis of transients). The solid lines show theoretical results from CAMRAD for the modal frequencies—with each curve labeled by the particular mode (regressing or progressing, flap or inplane, and body pitch and roll). The top set of curves correlates the data against CAMRAD (run without dynamic inflow), and the bottom set correlates the same data against CAMRAD (run with dynamic inflow).

In either set of plots, the nonrotating frequencies at $\Omega = 0$ emerge and begin to shift and couple as rpm is increased. Often, two modes couple together and then reemerge as new modal branches. In the top set of curves, the comparison of frequencies is good at lower rpm but begins to degrade as the rotor speed (and aerodynamic effectiveness) is increased. For Ω greater than 400 rpm, the predicted frequencies for the regressing

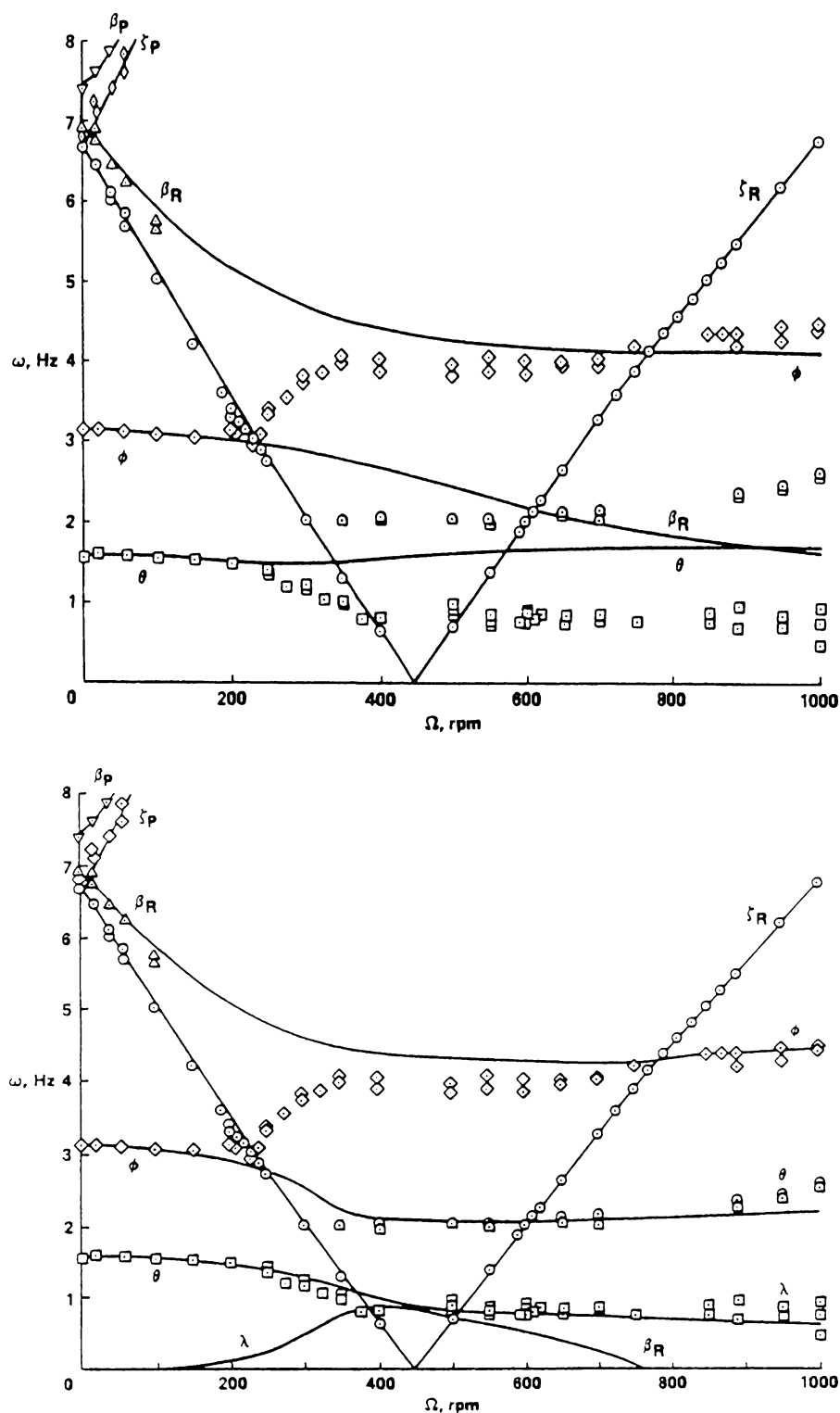


Fig. 10. Measured (—) and computed ($\circ$, $\square$) frequencies without (top curve) and with (bottom curve) dynamic inflow.

flap mode (β_R) and of the body pitch mode (θ) are not supported by the data, and the frequency of the body roll mode (ϕ) is slightly in error. Furthermore, the theory is predicting five branches, whereas the data display only four.

However, in the lower set of curves—with dynamic wake—theoretical curves pass directly through all four data branches. There is another especially interesting aspect of this correlation. In particular, Johnson used

the eigenvectors to identify the character of the modes, since the experimental data do not include any means of doing so. One normally looks at the largest participation of any degree of freedom in an eigenvector and thus identifies the mode with that motion. When Johnson looked at the frequency branches from CAMRAD (with the presence of dynamic inflow), he noticed that the regressing flap mode had become critically damped (0 frequency) and therefore had gone off of the plot.

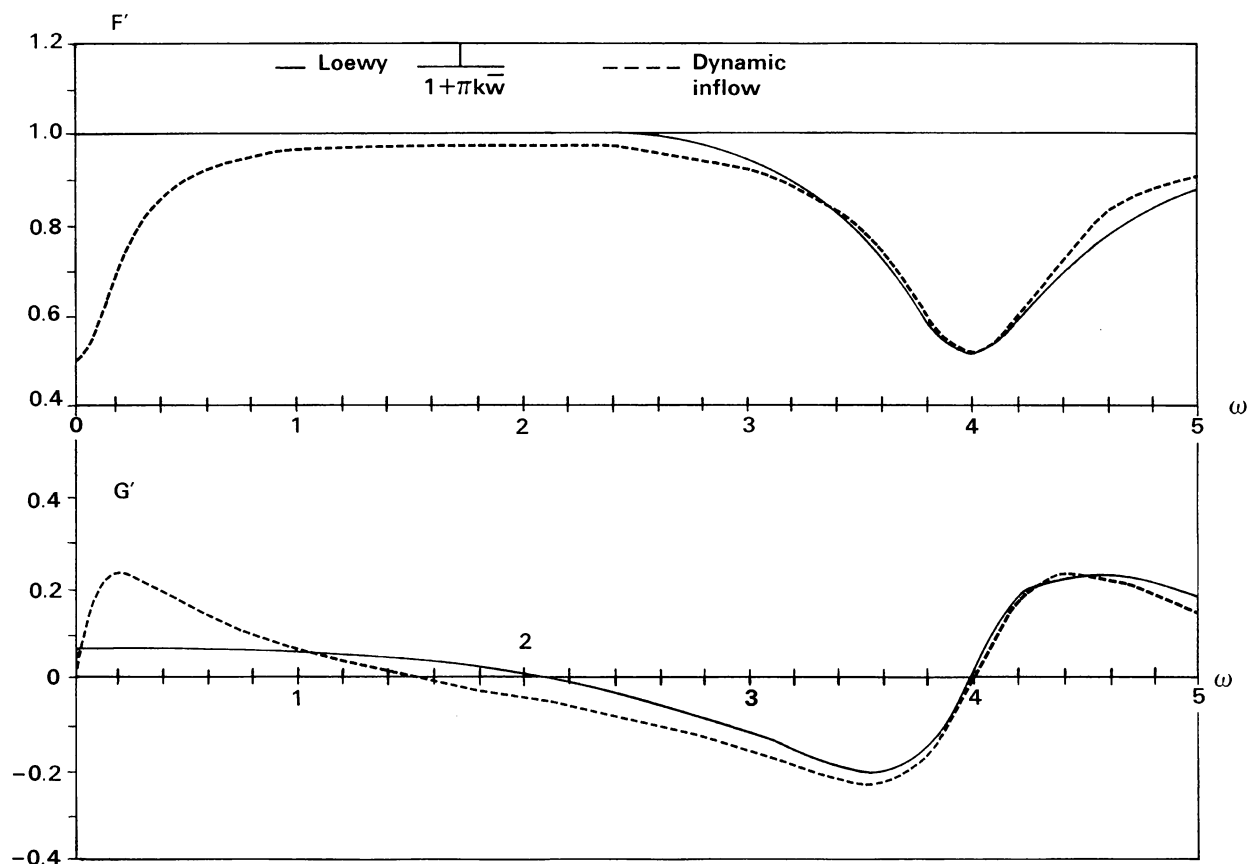


Fig. 11. Near-wake approximation to Loewy theory.

This explained the existence of only four modal frequencies in the data. In addition, Johnson noted that the eigenvector of the lowest frequency mode was dominated by the inflow states themselves rather than by any structural degree of freedom. Thus, he labeled that branch λ or “the inflow mode.” This was further confirmation of the importance of dynamic inflow to any rotor stability analysis.

Generalized theory

As it became clear that dynamic wake modeling was necessary for correct aeroelastic and stability modeling, more and more researchers began incorporating various inflow models into their codes. It was at this time that Friedmann at University of California, Los Angeles, began to use Padé approximants to turn traditional frequency-domain theories (such as Theodorsen and Loewy) into finite-state models that might compete with dynamic inflow (Ref. 18). As Friedmann applied his Padé approximation of Loewy theory to blade dynamics, he noted that there seemed to be large discrepancy between the results with Loewy theory and those with dynamic inflow. The discrepancy appeared especially pronounced at low frequencies for the collective mode.

Peters consequently turned his attention to understanding this apparent anomaly. As it turns out, the “discrepancy” was simply the result of a singularity in the Loewy theory for the collective mode at $\omega = 0$. This singularity had not been heretofore studied, but it explained the anomalies. However, as Peters was looking into the relationships between the Pitt and Loewy models, it occurred to him that it should be possible to extend dynamic inflow to include all harmonics and all radial distribu-

tions of inflow and thereby encompass Loewy theory as well as Sissingh theory in a single model.

A simple preliminary calculation showed that the dips in the Loewy function at integer-multiple frequencies could be represented through an extension of the potential functions used by Pitt. Figure 11 shows the approximate comparison that was created. As Peters looked at the comparison of the imaginary part of the Loewy function with the Pitt theory, he further realized that there was a relationship between the time constants of the various harmonics as predicted by the two respective models (i.e., the slope of the imaginary part G' at these integers. In particular, one could obtain simple, closed-form representations of the time constant for the lowest mode of the m th harmonic from either representation.

$$\begin{aligned} \text{Dynamic wake: } T &= 0.75/(m + 3/2) \\ \text{Loewy at } r = 3/4: T &= 0.75/m \end{aligned} \quad (12)$$

Equation (12) shows that the time constants of the two theories are very similar (especially for harmonics of $m = 4$ and above. In contrast, the $m = 1$ time constant from Loewy theory ($T = 0.75$) is 2.5 times as large as is the true value from Pitt ($T = 0.30$), and the discrepancy is even greater for $m = 0$ in which case Loewy gives $T = \infty$ rather than $T = 0.50$ (and this accounts for the singularity noted by Peters). Of course, for $m = 0$ and $m = 1$, the dynamic wake model is known to have the correct time constants because of comparisons both with Carpenter and Fridovitch and with the NASA data. Therefore, Eq. (12) implies that a more general theory (based on potential flow theory) might well give both the Pitt model and an improvement to Loewy theory.

Peters took these preliminary comparisons to Bob Ormiston and Bill Warmbrodt in their respective Army and NASA groups and convinced them to give him some funding to pursue this line of extension of dynamic wake models. The funding arrived in 1985, just as Peters moved from Washington University to Georgia Tech, where he joined the Rotorcraft Center of Excellence. Robin Gray (the 1991 Nikolsky Lecturer) had successfully attracted funding for the Rotorcraft Center in 1982. (Of note is the fact that Gray had worked under Alexander Nikolsky on rotor inflow (Ref. 19).) By 1985, one of Peters' former doctoral students, Dan Schrage (the 1999 Nikolsky Lecturer), had come as the new director of that Center.

Peters' first doctoral student at Georgia Tech was Cheng Jian He, who had worked on wake modeling under Wang Shi Cun in China. Peters put He on the task of extending the Pitt model to be completely general. Together, Peters and He were able to derive (from the three-dimensional potential flow equations) an entirely closed form, general version of the dynamic wake model. This new model includes all harmonics and all radial distributions of inflow at arbitrary wake skew angle (ranging from axial flow to edgewise flow) (Ref. 20). The theory is in closed form and is amazingly compact—such that everything needed to code the method can be expressed in a 10-point font on a single side of an 8.5×11 sheet of paper. In addition, the He model contains the Pitt model as a special case (when the number of states is truncated to 3). This makes it very convenient to upgrade existing codes from the Pitt model to the He model.

At this same time, some excellent laser velocimeter data were being taken in the NASA Langley Transonic Dynamics Tunnel. These gave the induced flow field of a rotor in forward flight. Data were taken for two different blade sets, for various advance ratios, and for various shaft angles. It was the perfect venue for He to test out the new theory (Ref. 21). Shown in Fig. 12 is the inflow comparison for the tapered blades at an advance ratio of 0.15. The correlation is similarly good at all other measured conditions.

Q4 Also shown in Fig. 12 are data from both a prescribed-wake model and a free-wake model. Incredibly, the He model gives as good or better correlation than is provided by the vortex-lattice methods. The reason for this is fairly straightforward. Although the vortex-lattice models have more physics than does the potential flow model, the He model has just enough physics to compute the flow—but no more. Also, whereas vortex-lattice models have convergence issues—with the number of trailing filaments, number of shed filaments, the number of turns of the wake, and the vortex core size—the He model is robust and based on orthogonal

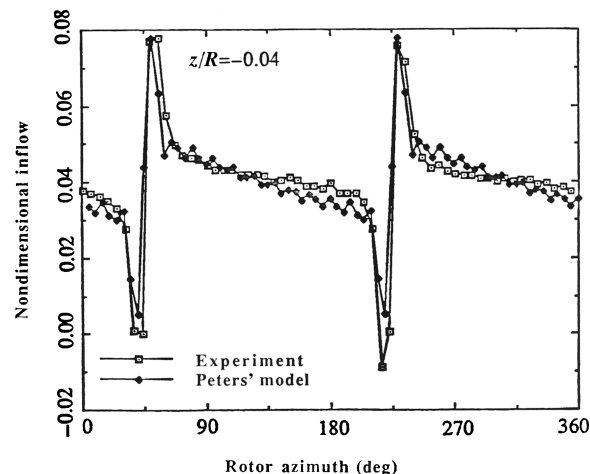


Fig. 13. Inflow for two-bladed rotor in Hover from Georgia Tech Hover stand.

potential functions. Thus, it is always at the optimum convergence for any number of states. That is why it fares so well in comparison.

Another of Peters' students, Ay Su, soon began work on how to nonlinearize the wake model so that it would be applicable even in hover. Su showed that the model could be made nonlinear in the uniform-flow state in such a way that it would agree with He in forward flight and with nonlinear momentum theory in hover (Ref. 22). Fortunately, Prof. Narayanan Komerath of Georgia Tech concurrently produced laser data from the new Georgia Tech Hover Test stand. Correlations between the nonlinear version of the theory and the hover experimental data, as given in Ref. 21, are excellent. For example, Fig. 13 shows the unsteady inflow at a fixed point in space for a two-bladed model rotor. Interestingly, there at first was a phase discrepancy between theory and experiment. However, rather than being attributed to the theory, it turned out to be due to an error in the phasing of the experimental data collection system.

Another noteworthy development at this time was that one of Dr. Schrage's students, Bill Lewis, did a Ph.D. thesis on how pilots perceived the performance of flight simulators when the simulators had various fidelity levels of dynamic wake models included (Ref. 23). This authenticated that pilots could tell the improvement in the realistic nature of flight simulation when dynamic inflow was modeled correctly.

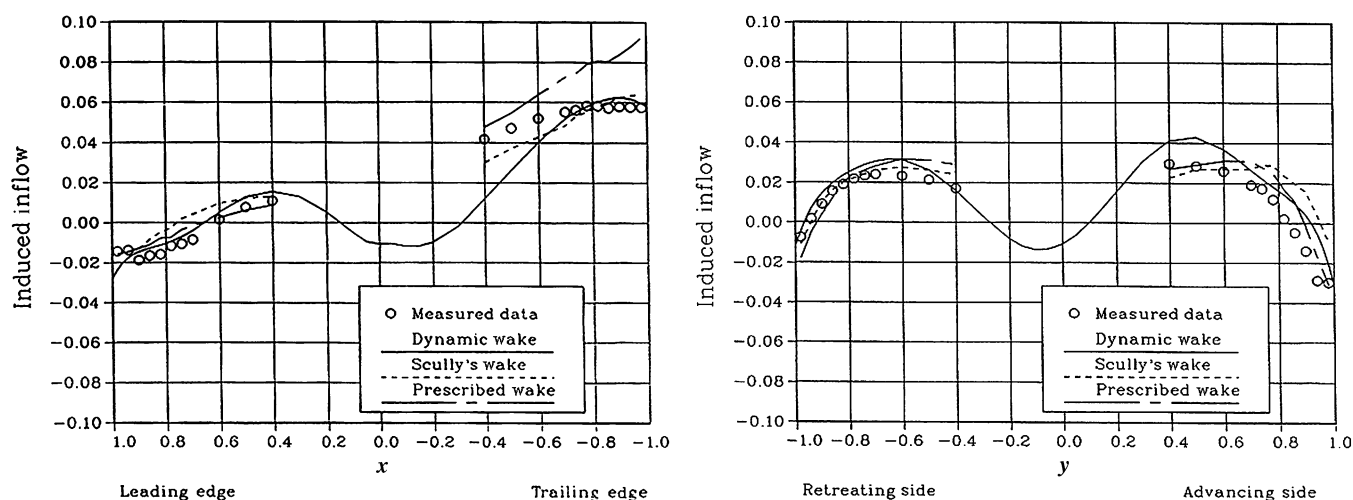


Fig. 12. Inflow correlation at $\mu = 0.15$.

Thus, the He model was established as a generalized theory that not only had good validation in hover and forward flight, but that also reproduced the Pitt model as a special case. After his Ph.D., Cheng Jian He stayed on at Georgia Tech for some time as a postdoctoral fellow. He later joined Advanced Rotorcraft Technology and began to put his model into flight simulation codes such as FLIGHTLAB. Soon the He model was also in the new comprehensive Army code, RCAS. Generalized dynamic wake models had arrived in the mainstream of aerodynamic applications.

Refinement (1990–2008)

The last two decades of dynamic wake modeling have seen a series of improvements made to the dynamic inflow and dynamic wake models. As was the case in the previous history, these new improvements were generally motivated by unexplained experimental data—followed by physically intuitive arguments by the researchers—then consummated by application of the simplest possible physics to put numbers to the ideas. The three main refinements have been (1) the addition of wake curvature, (2) the extension to all components of flow both on and off the rotor disk, and (3) the addition of a correction for swirl velocity.

Wake curvature

As the new wake models found their way into flight simulation codes, it became clear that there existed a phenomenon explained neither by dynamic inflow models nor by the vortex models. In particular, there was the issue of off-axis coupling. When a helicopter makes a pull-up maneuver via a cyclic stick input, there is inevitably an accompanying roll response—an off-axis response. However, the simulation codes consistently predicted the off-axis response to be of opposite sign from the flight test data.

Aviv Rosen postulated that the off-axis effect might be due to the fact that, during a pull-up, the wake distorts into a curved helix—thus stacking vortex layers more tightly together on one side than on the other (Ref. 24). The differential in vortex spacing could cause a differential in velocity that might explain the phenomenon. Rosen demonstrated that a simple vortex-wake model of the curvature could indeed explain part of the off-axis error. Keller and Curtiss further showed that this effect could alternatively be captured by simple momentum theory with a curved momentum tube (Ref. 25). They expressed the effect as a forcing function (proportional to rotor pitch rate) placed on the right-hand side of the dynamic inflow equations.

Based on this, Prasad at Georgia Tech (along with his students and Peters) extended the Pitt and He dynamic wake models to include terms due to wake curvature (the result of pitch rate). Prasad and his students moved the curvature terms to the left-hand side of the equations and placed them within the $[L]$ matrix. Thus, rather than appearing as forcing functions, they are treated along with wake skew and wake contraction as wake deformation effects on $[L]$ (Ref. 24). Figure 14 shows the three wake deformation variables that consequently appear in the equations via $[L]$. Wake contraction is included in the V parameter, as introduced by Pitt and generalized by Su. The wake skew appears in the X parameter, where X is tangent of one half of the skew angle. (Thus, X varies from 0 to 1.0 as skew goes from 0° in hover to 90° in edgewise flow.) Third, wake curvature appears as additional terms in the $[L]$ matrix due to either lateral or longitudinal curvature κ .

Prasad understood that, just as dynamic inflow incurs a time delay for the developing flow field, the three distortion parameters also require first-order time delays for their development. With the new formulation, Prasad proved that the off-axis coupling could be predicted (Fig. 15). The formulation of the new model implies that the previous dynamic

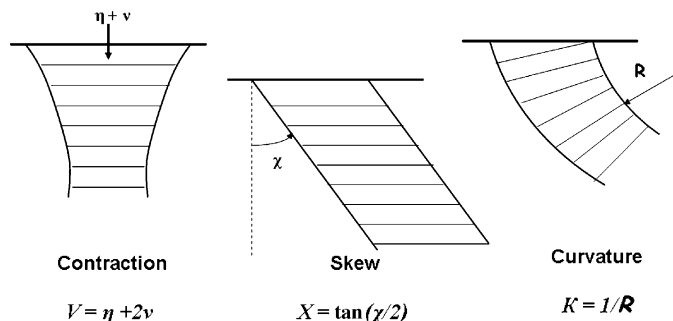


Fig. 14. Wake distortion parameters.

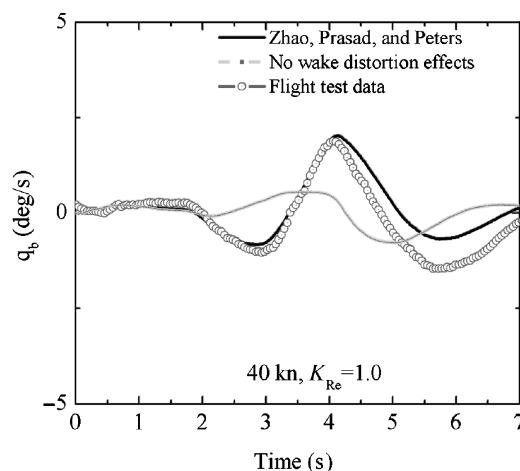


Fig. 15. Prediction of off-axis coupling of UH-60.

wake theories are included in the new theory (as the special case when curvature is zero). Thus, the wake models remain hierarchical.

It is interesting to note that early applications of wake curvature (e.g., Refs. 22, 23) needed to add a magnification factor K_{Re} to the wake-curvature effect to obtain good correlation. Often the factor had to be as large as 4.0 or more. The results in Ref. 24 and Fig. 15, however, show that—when all of the effects of nonuniform flow and tail interactions are included, the magnitude of the wake-curvature effect from momentum theory ($K_{Re} = 1.0$) gives good correlation.

Off-Rotor Flow Field

Another set of experimental data that was causing concern at this time was a set of data in ground effect taken by the Japanese. These data involved both a dynamically moving ground plan and a partial ground plane (such as a rotor hovering partly over a ship deck). To handle this type of behavior, investigators decided that perhaps an actuator disk, such as is modeled in dynamic inflow theory, might be put at the ground plane. Such a disk could be translated and rotated—to simulate ship deck motion—or could be partially loaded to simulate partial ground.

To accommodate this approach, the actuator disk would have to include mass sources (to simulate the nonpenetration boundary condition), and one would need to be able to compute all three components of induced flow both on and above that disk. Peters' students Morillo, Yu, and Hsieh set out to generalize the model in just this way.

Morillo showed that such an inflow theory could be derived directly from the potential flow equations via a Galerkin approach in which the comparison functions are not velocity fields at the rotor disk (as He had assumed) but are velocity potentials (Ref. 27). Morillo included

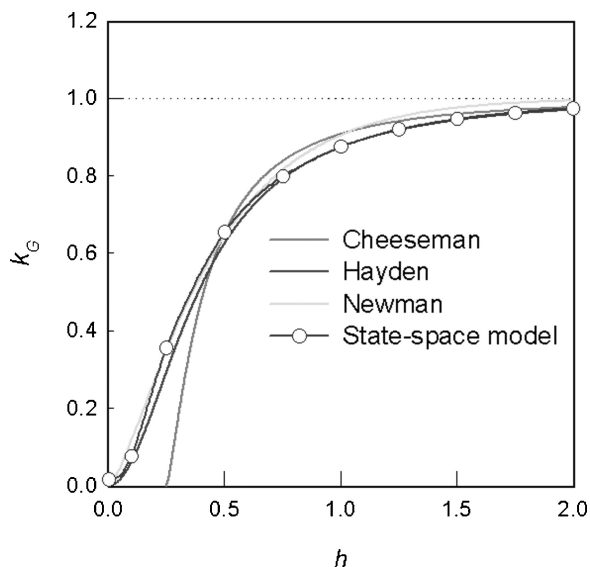


Fig. 16. Ground effect by dynamic wake model.

all admissible potential functions—including those that represent mass sources at the disk as well as those that represent pressure discontinuities. The result of application of the Galerkin method is a theory that includes strengths of mass sources on the rotor as forcing functions. Since the

states are velocity potentials, all three components of the entire flow field can be known. Morillo successfully derived this theory in a compact closed form (with the exception of some singular potentials). The theory included the He theory as the special case for which the potential functions with mass sources are truncated. Thus, once again, previous theories are hierarchical to this one.

Yu (Ref. 26) applied the theory to ground effect, showing that it could indeed predict the effect of moving, tilted, and partial ground planes, as well as a standard ground plane, as shown in Fig. 16, which gives the ground effect factor K_G as a function of rotor height in rotor radii. To complete the development, Hsieh added a closed-form version of the singular velocity potentials associated with some of the mass-injection terms (Ref. 29), for which Morillo had been unable to obtain a closed form. Thus, the theory became complete for all components of flow both on and off the disk.

Swirl velocity

The final refinement to be discussed here is that of swirl velocity. The need for this arose from an unlikely source. Frank Harris (the 2006 Nikolsky Lecturer) had shown that the measured induced power of a rotor was far greater than the minimum possible induced power predicted by Glauert. The burning question was why we were not designing rotors anywhere near this optimum power. Bob Ormiston suggested that dynamic wake models would be perfect to do such a study of optimization (due

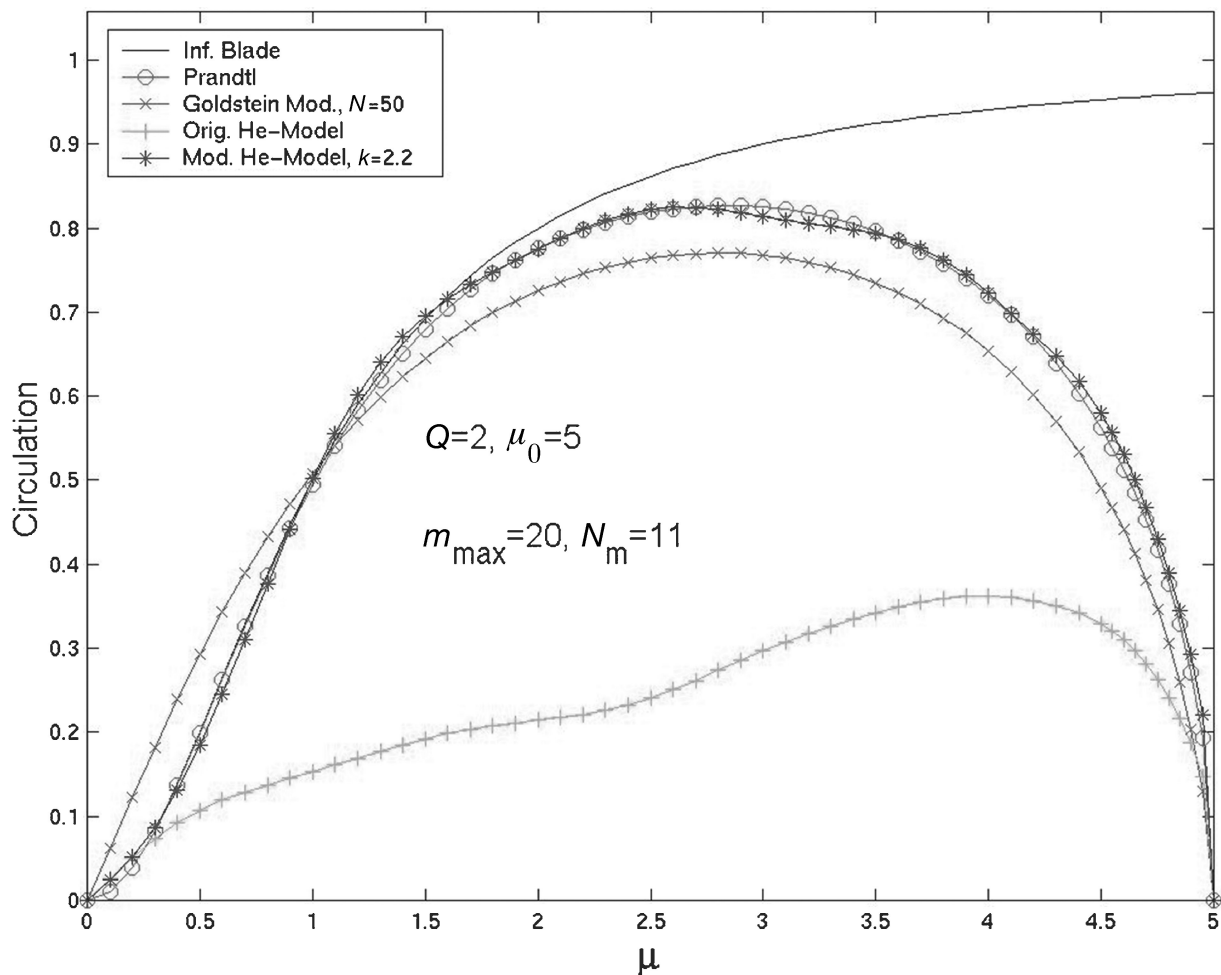


Fig. 17. Effect of swirl on optimum circulation: high climb rate.

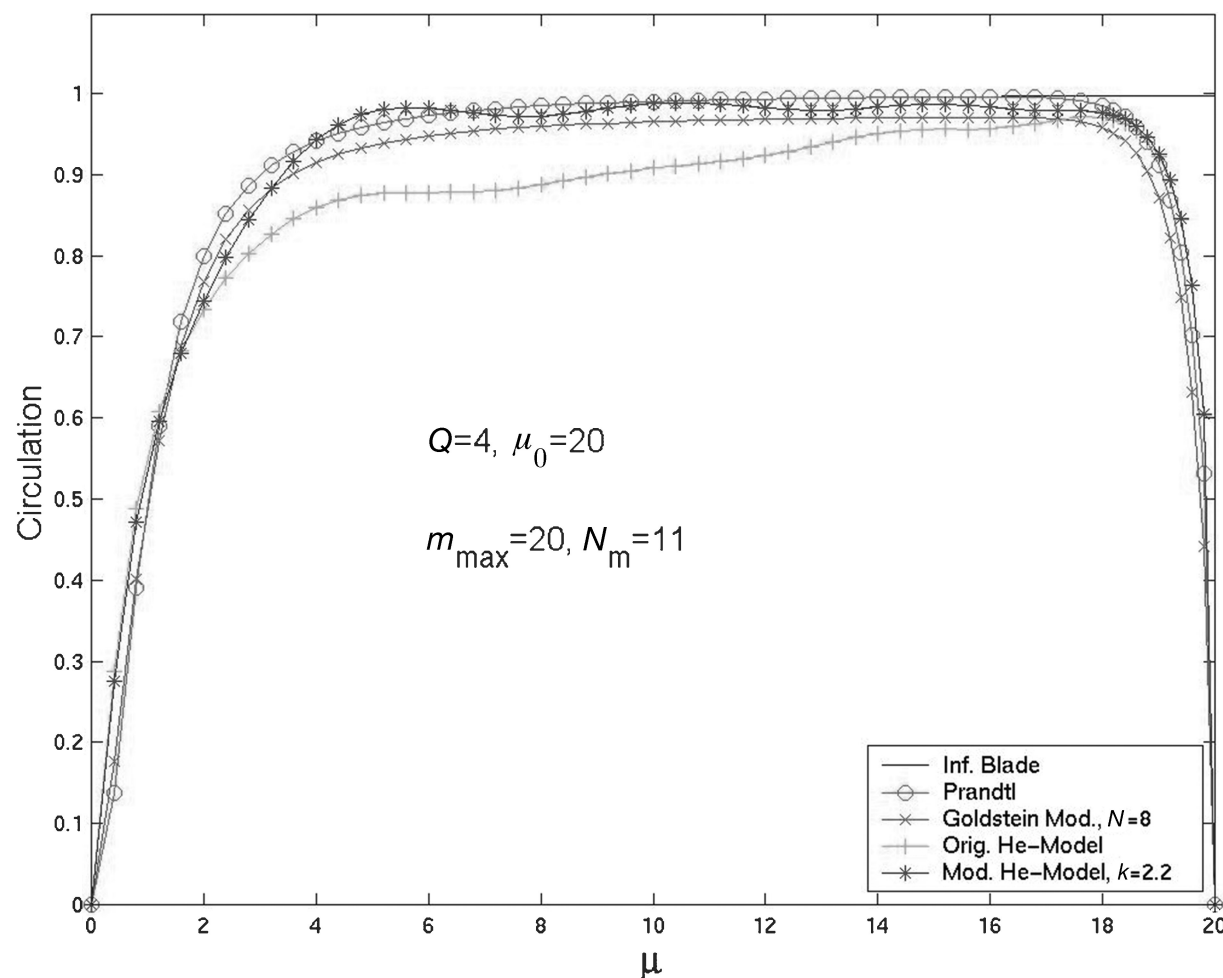


Fig. 18. Effect of swirl on optimum circulation: low climb rate.

to their matrix form). Dynamic wake models, however, are actuator disk theories—which implies an assumption that the rotor lift is normal to the disk. Thus, no swirl energy is added to the wake. The kinetic energy lost to swirl is necessary if one is to do an adequate power optimization at high inflow angles.

Since finite-state dynamic wake models can be derived from Ritz–Galerkin methods, it seemed logical that one could add the kinetic energy due to swirl, then do the Galerkin analysis, and thereby generalize the model. Makinen (Ref. 30) took on this task and was able to derive a simple correction to the mass matrix of dynamic wake models based on swirl energy. Figures 17 and 18 show comparisons of the bound circulation for an optimum propeller at two climb rates $\mu_0 = 1/\lambda$. One is a very steep ascent $\mu_0 = 5$ and the other is a slower climb $\mu_0 = 20$. In these plots, $\mu = r/\lambda$ and not the advance ratio. Also shown in the figures are the exact result from Goldstein and the approximate optimum result from Prandtl. The addition of the Makinen swirl correction adequately captures the effect of the wake swirl and allows dynamic wake models to be used for power optimization. In part, results of a formal quadratic optimization (with swirl terms) gives the same optimum circulation distribution as in the classical theories.

Future refinements

There is no doubt that, as time goes by, other refinements will be made to dynamic wake models, and it is expected that these will simi-

larly be driven by the need to model experimentally determined behavior of the rotor inflow. However, such refinements will never bring dynamic inflow or dynamic wake models to the point at which they replace CFD or vortex–lattice codes. The niche for dynamic inflow models will probably remain that of handling qualities, aeroelasticity, real-time flight simulation, and preliminary design optimization. These are the areas in which necessity of reasonable computer time and the need for physical insight dominate the user needs over requirements for comprehensive capabilities.

Summary and Conclusions

Through the years, dynamic inflow and dynamic wake models have proven competitive and have thusly survived the emergence of more powerful tools such as CFD and vortex–lattice models. The reasons for this survival are fairly straightforward.

- 1) The models have been derived in response to experimental data.
- 2) They have enough texture to explain the critical phenomenon—but no more.
- 3) They are hierarchical so that previous models are always contained in later.
- 4) What they lack in detail, they make up in robust and speedy convergence.

These factors have kept dynamic inflow models at the forefront of rotorcraft analysis.

Despite the inroads that other models will make as computers become more powerful, I do not believe that dynamic wake models (and other simplified rotorcraft analysis tools) will ever become obsolete. There are three reasons for this. First, no matter how fast computers become, they can never do everything. Thus, simplified models will always find a place. Second, there will always be a need for real-time simulation. Thus, there will always be a need for simplified models that can execute in reasonable CPU times. Third, simple models—such as dynamic inflow—give direct intuitive connections among the equations, the physics, and the results. Thus, they are indispensable for gaining insight into rotorcraft behavior. Simplified models will always be with us.

Acknowledgments

I would like, first of all, to thank Kurt Hohenemser, who introduced me to the ideas of helicopter dynamics and aerodynamics back in 1968. In doing the research for this paper, I came across a copy of the book by Alexander Nikolsky, *Helicopter Analysis*, John Wiley & Sons, New York, 1951. That book was given to me by Kurt shortly before his death, and it includes the following inscription from Nikolsky to Hohenemser:

To Dr. Kurt H. Hohenemser with recognition of his pioneering in stability work on Helicopters from A. A. Nikolsky.

This autograph is a fitting connection as I acknowledge in this Nikolsky lecture the fundamental influence that Kurt had on my life.

I would also like to acknowledge Bob Ormiston, the best boss I ever had. When I first came to the Army Air Mobility Research and Development Laboratory in 1970, Bob told me, “Dave, some day you will bring me a curve that has a glitch in it. At that point, I’m going to say, ‘Dave, what is that glitch?’ If your answer to me is, ‘I don’t know; that’s the way it came out of the computer,’ you’re fired.” That was not the only great piece of advice that Bob gave me, but it was one of the best.

Third, I would like to thank Dewey Hodges, one of my greatest friends, who always kidded that structures and dynamics were the “exact sciences” whereas aerodynamics was an “inexact science.” Actually, the whole history of dynamic inflow models can be characterized as models derived by we dynamicists looking at aerodynamics from our point of view. Dynamic wake models are in fact applied aerodynamics, not pure aerodynamics—and that has colored the way the theory has developed.

I would like to thank the U.S. Army Research Office, the U.S. Army Aeroflightdynamics and Technology Directorate, NASA Ames Research Center, and the Georgia Tech Center of Excellence for Rotary-Wing Aircraft Technology for sponsoring the various aspects of this work through the years.

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Author's queries:

Q1: Provide caption for Figure 2.

Q2: No appendix is given in the article. Is Appendix A a part of Ref. 6?

Q3: Confirm whether Army Aeronautical Research Laboratory is abbreviated correctly.

Q4: Confirm whether citation of Fig. 12 is OK here. If not, cite it at the appropriate place in the text.

Q5: At Ref. 26, Zhao et al. is cited in the References not Yu. Should Ref. 28 be cited in place of Ref. 26?

Q6: Check whether the term “particular” is meant here instead of “partiche.”

Q7: Provide inclusive dates of the proceedings.